

Jerrold E. Marsden and Anthony J. Tromba

Vector Calculus

Fifth Edition

Chapter 7:

Integrals over Paths and Surfaces

7.1 The Path Integral

Copyright © 2003 by W. H. Freeman & Company

7.1 The Path Integral

Key Points in this Section.

1. **Definition.** The *path integral* of a scalar function f in $\mathbb{R}^3$ along a path $\mathbf{c}(t)$, where $a \leq t \leq b$, is defined by

$$\int_{\mathbf{c}} f \, ds = \int_a^b f(x(t), y(t), z(t)) \|\mathbf{c}'(t)\| \, dt.$$

2. The *scalar element of arc length* is

$$ds = \|\mathbf{c}'(t)\| \, dt.$$

3. There is a similar definition for path integrals in the plane (just leave out the z -dependence).
4. If f has the interpretation of the mass density along a wire, then the path integral is the total mass of the wire.

5. If the curve is in the xy -plane and f is interpreted as the height of a fence along the curve, then the path integral is the area of (one side of) this fence.
6. **Arc Length a Special Case.** If $f = 1$ (is identically one), then the definition of the path integral reduces to that for the arc length of the path.

Integral de línia de funció escalar

DEFINITION: Path Integrals The *path integral*, or the *integral of* $f(x, y, z)$ *along the path* $\mathbf{c}$, is defined when $\mathbf{c}: I = [a, b] \rightarrow \mathbb{R}^3$ is of class C^1 and when the composite function $t \mapsto f(x(t), y(t), z(t))$ is continuous on I . We define this integral by the equation

$$\int_{\mathbf{c}} f \, ds = \int_a^b f(x(t), y(t), z(t)) \|\mathbf{c}'(t)\| \, dt.$$

Sometimes $\int_{\mathbf{c}} f \, ds$ is denoted

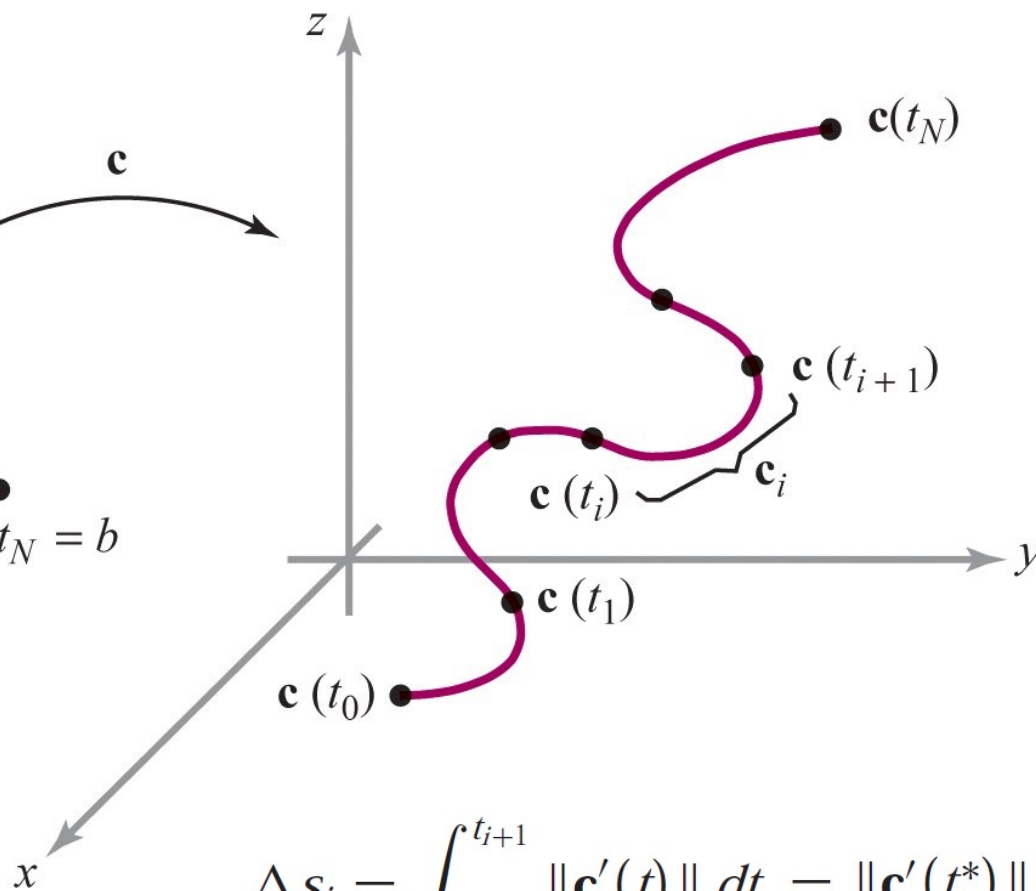
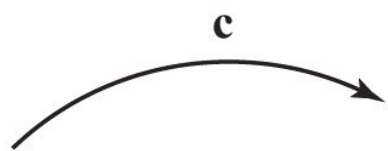
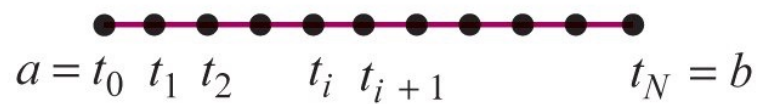
$$\int_{\mathbf{c}} f(x, y, z) \, ds$$

or

$$\int_a^b f(\mathbf{c}(t)) \|\mathbf{c}'(t)\| \, dt.$$

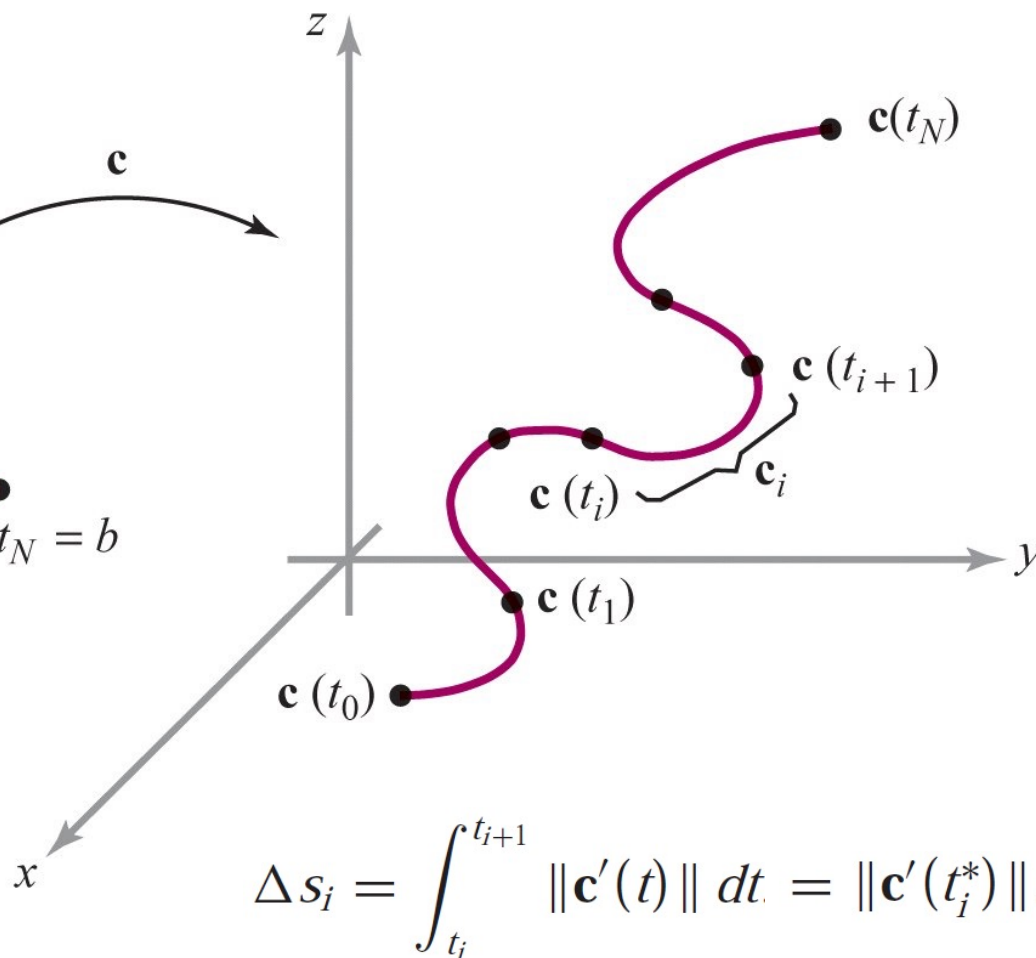
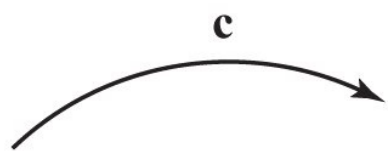
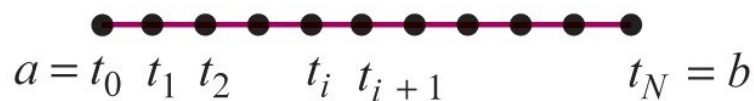
If $\mathbf{c}(t)$ is only piecewise C^1 or $f(\mathbf{c}(t))$ is piecewise continuous, we define $\int_{\mathbf{c}} f \, ds$ by breaking $[a, b]$ into pieces over which $f(\mathbf{c}(t)) \|\mathbf{c}'(t)\|$ is continuous, and summing the integrals over the pieces.

$$S_N = \sum_{i=0}^{N-1} f(x_i, y_i, z_i) \Delta s_i$$



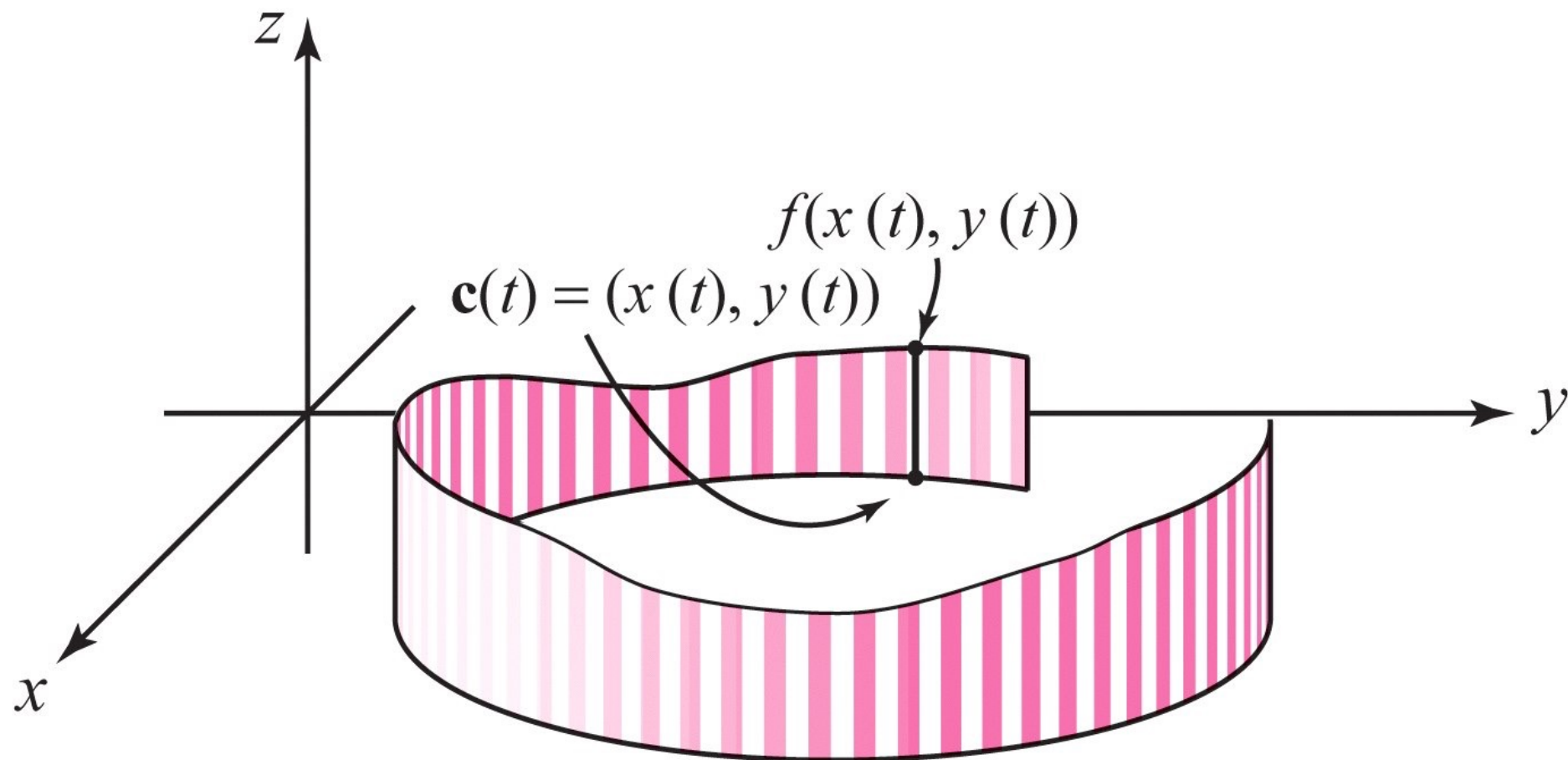
$$\Delta s_i = \int_{t_i}^{t_{i+1}} \|\mathbf{c}'(t)\| dt = \|\mathbf{c}'(t_i^*)\| \Delta t_i$$

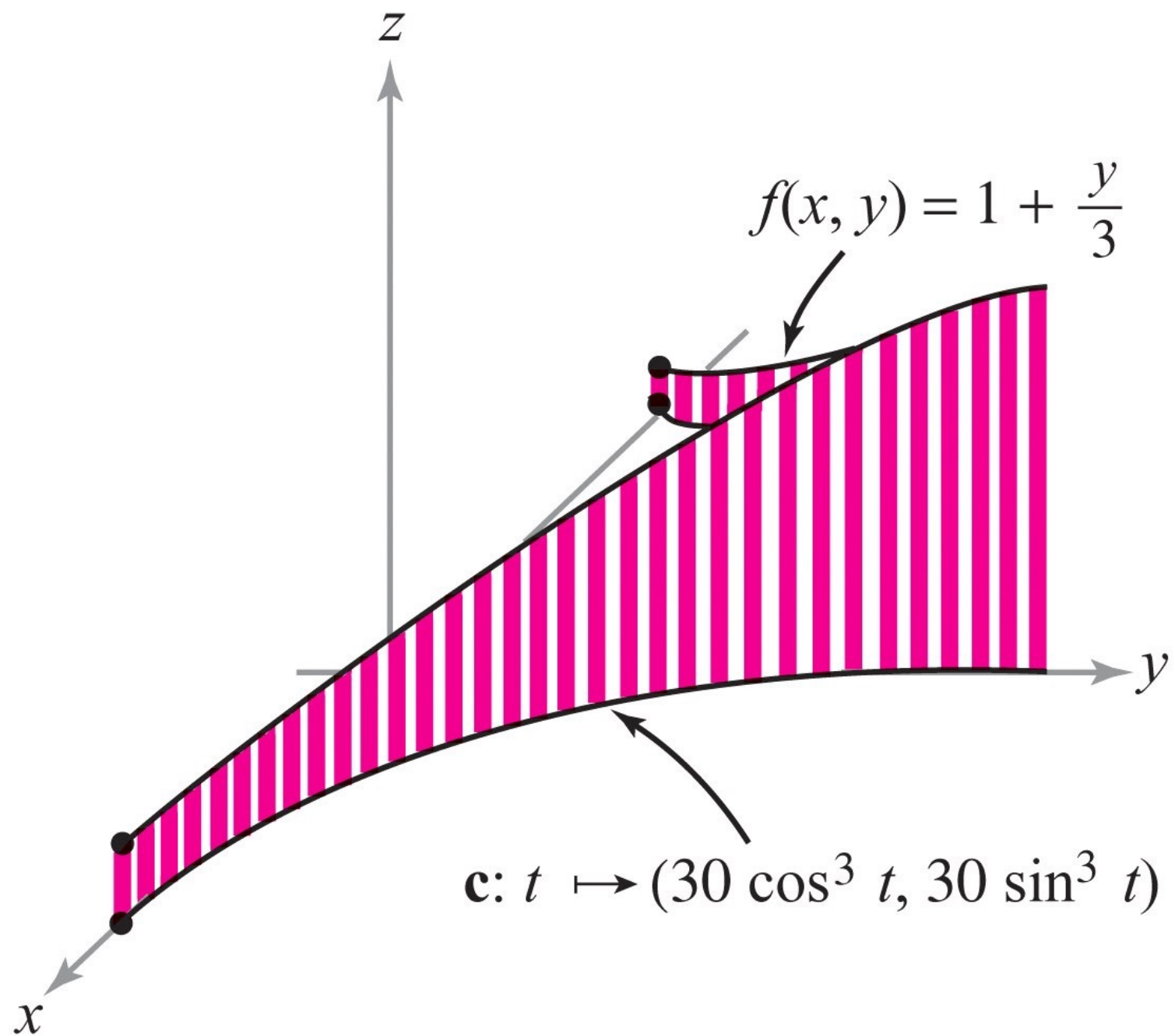
$$S_N = \sum_{i=0}^{N-1} f(x_i, y_i, z_i) \Delta s_i$$

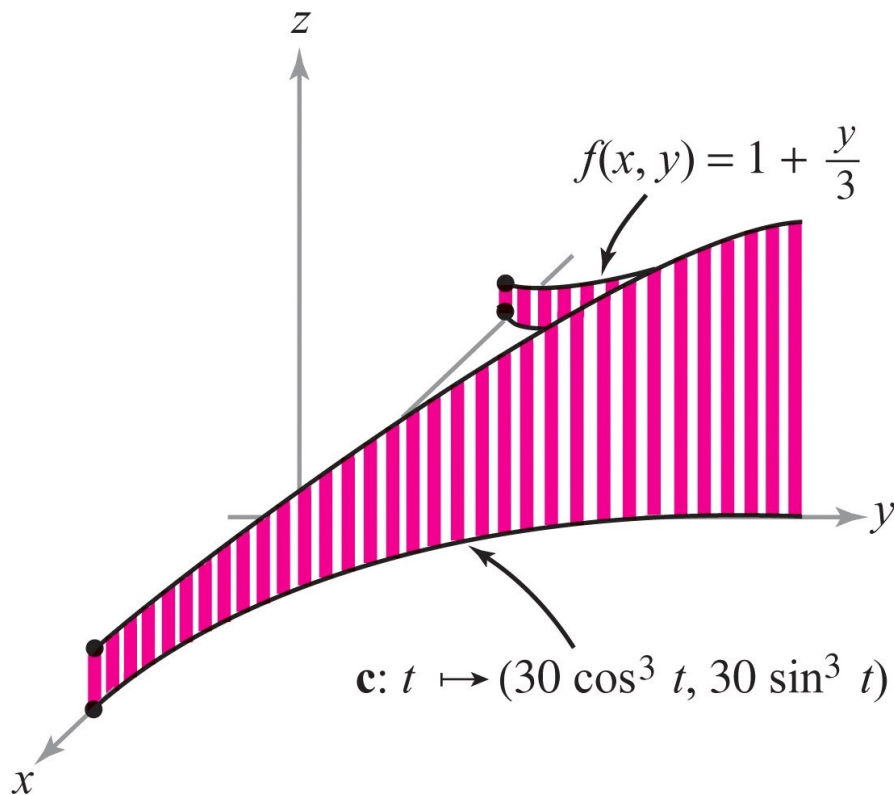


$$\Delta s_i = \int_{t_i}^{t_{i+1}} \|\mathbf{c}'(t)\| dt = \|\mathbf{c}'(t_i^*)\| \Delta t_i$$

$$\begin{aligned} \int_{\mathbf{c}} f(x, y, z) ds &= \lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} \sum_{i=0}^{N-1} f(x_i, y_i, z_i) \|\mathbf{c}'(t_i^*)\| \Delta t_i \\ &= \int_I f(x(t), y(t), z(t)) \|\mathbf{c}'(t)\| dt \end{aligned}$$







Superfície al primer quadrant

$$\mathbf{c}'(t) = (-90 \cos^2 t \sin t, 90 \sin^2 t \cos t)$$

$$\|\mathbf{c}'(t)\| = 90 \sin t \cos t$$

$$\int_{\mathbf{c}} \left(1 + \frac{y}{3}\right) ds = \int_0^{\pi/2} \left(1 + \frac{30 \sin^3 t}{3}\right) 90 \sin t \cos t dt$$

$$= 90 \int_0^{\pi/2} (\sin t + 10 \sin^4 t) \cos t dt$$

$$= 90 \left[\frac{\sin^2 t}{2} + 2 \sin^5 t \right]_0^{\pi/2} = 90 \left(\frac{1}{2} + 2 \right) = 225$$

Jerrold E. Marsden and Anthony J. Tromba

Vector Calculus

Fifth Edition

Chapter 7:

Integrals over Paths and Surfaces

7.2 Line Integrals

7.2 Line Integrals

Key Points in this Section.

1. **Definition.** The *line integral* of a given continuous vector field $\mathbf{F}$ (defined in the plane or in space) along a path $\mathbf{c}(t)$, where $a \leq t \leq b$, is defined by

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t) dt.$$

2. The *vector line element* is

$$d\mathbf{s} = \mathbf{c}'(t) dt.$$

3. **Interpretation as Work.** If $\mathbf{F}$ represents a force field, then the line integral of $\mathbf{F}$ along $\mathbf{c}$ is the *work done* by the force field in moving a particle subject to this force field, along the path. (Another interpretation in terms of circulation, when $\mathbf{F}$ represents the velocity field of a fluid, is given in Chapter 8).

4. **Line Integral of a Gradient.** If $\mathbf{F} = \nabla f$, then an analog of the *Fundamental Theorem of Calculus* holds

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = f(\mathbf{c}(b)) - f(\mathbf{c}(a)).$$

In fact, this result follows directly from the single variable Fundamental Theorem of Calculus since, by the Chain Rule,

$$\frac{d}{dt} f(\mathbf{c}(t)) = \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t).$$

5. Line integrals are independent of orientation preserving reparametrizations and path integrals are independent of *any* reparametrization. This is proved using the single-variable change of variables formula.

6. Because of the independence of parametrization one can define the line integral of a vector field along a geometric curve C , denoted

$$\int_C \mathbf{F} \cdot d\mathbf{s} \quad \text{or} \quad \int_C \mathbf{F} \cdot d\mathbf{r}$$

as long as an orientation along the curve is specified. To actually evaluate such an integral, any parametrization may be chosen, or some other method (such as the fundamental theorem in item 3) is used.

Integral de línia de funció vectorial

DEFINITION: Line Integrals Let $\mathbf{F}$ be a vector field on $\mathbb{R}^3$ that is continuous on the C^1 path $\mathbf{c}: [a, b] \rightarrow \mathbb{R}^3$. We define $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$, the *line integral* of $\mathbf{F}$ along $\mathbf{c}$, by the formula

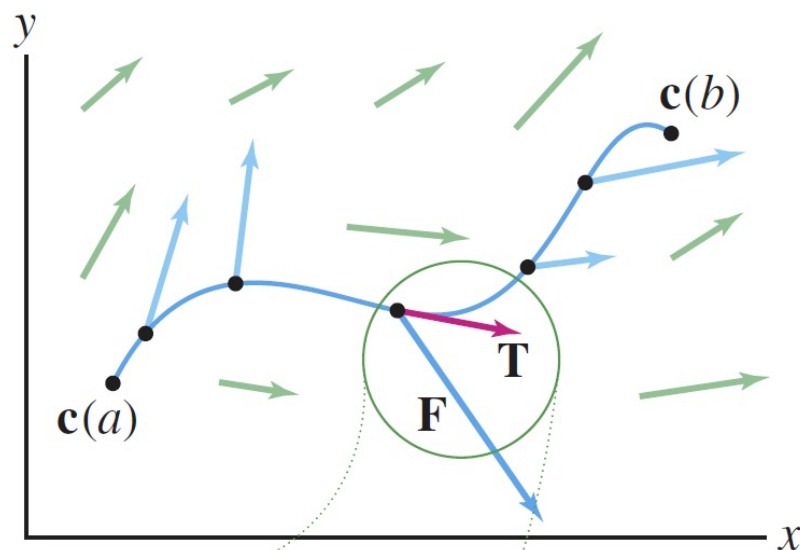
$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t) dt;$$

that is, we integrate the dot product of $\mathbf{F}$ with $\mathbf{c}'$ over the interval $[a, b]$.

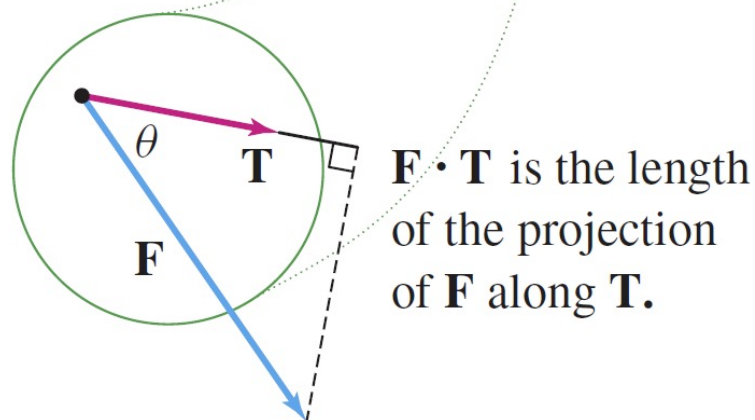
As is the case with scalar functions, we can also define $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ if $\mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)$ is only piecewise continuous.

DEFINITION Vector Line Integral The line integral of a vector field $\mathbf{F}$ along an oriented curve $\mathcal{C}$ is the integral of the tangential component of $\mathbf{F}$:

$$\int_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathcal{C}} (\mathbf{F} \cdot \mathbf{T}) ds$$



$\mathbf{T}(P)$ vector **tangent unitari** al punt P de la corba $\mathbf{c}$



$$\mathbf{F}(P) \cdot \mathbf{T}(P) = \|\mathbf{F}(P)\| \|\mathbf{T}(P)\| \cos \theta = \|\mathbf{F}(P)\| \cos \theta$$

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_C (\mathbf{F} \cdot \mathbf{T}) ds$$

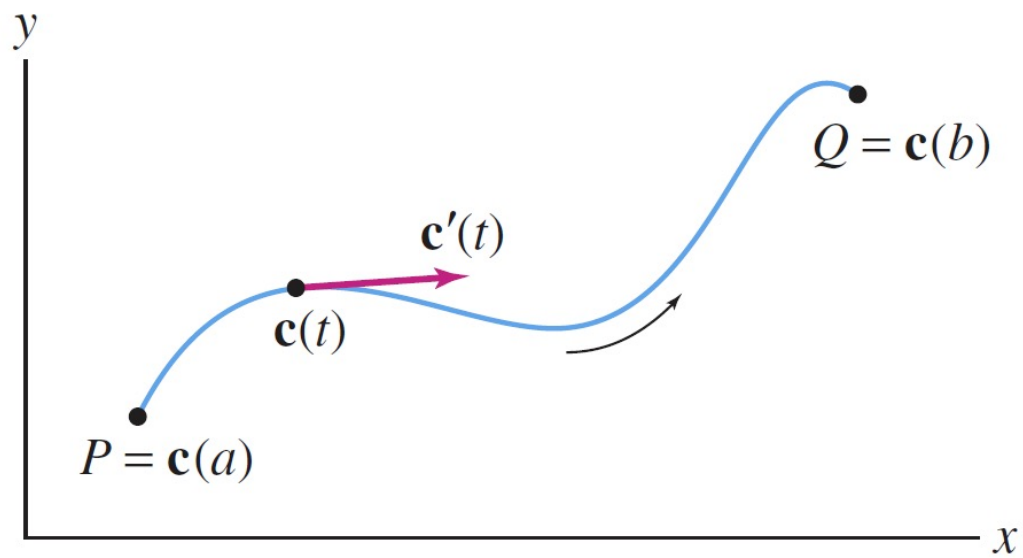
$\mathbf{T}(P)$ vector **tangent unitari** al punt P de la corba $\mathbf{c}$ $\mathbf{T} = \frac{\mathbf{c}'(t)}{\|\mathbf{c}'(t)\|}$

$$(\mathbf{F} \cdot \mathbf{T}) ds = \left(\mathbf{F}(\mathbf{c}(t)) \cdot \frac{\mathbf{c}'(t)}{\|\mathbf{c}'(t)\|} \right) \|\mathbf{c}'(t)\| dt = \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t) dt$$

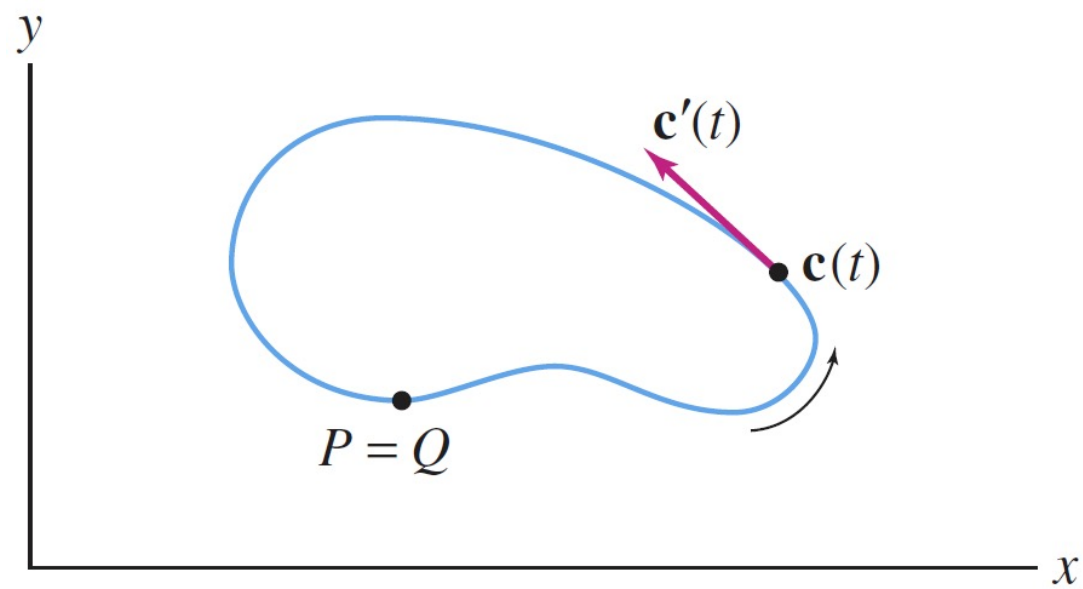
THEOREM 2 Computing a Vector Line Integral If $\mathbf{c}(t)$ is a regular parametrization of an oriented curve C for $a \leq t \leq b$, then

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t) dt$$

Simbòlicament: $ds = \|\mathbf{c}'(t)\| dt$



(A) Oriented path from P to Q



(B) A closed oriented path

Notació i càlcul

Si

$$\mathbf{F}(x, y, z) = (F_1(x, y, z), F_2(x, y, z), F_3(x, y, z))$$

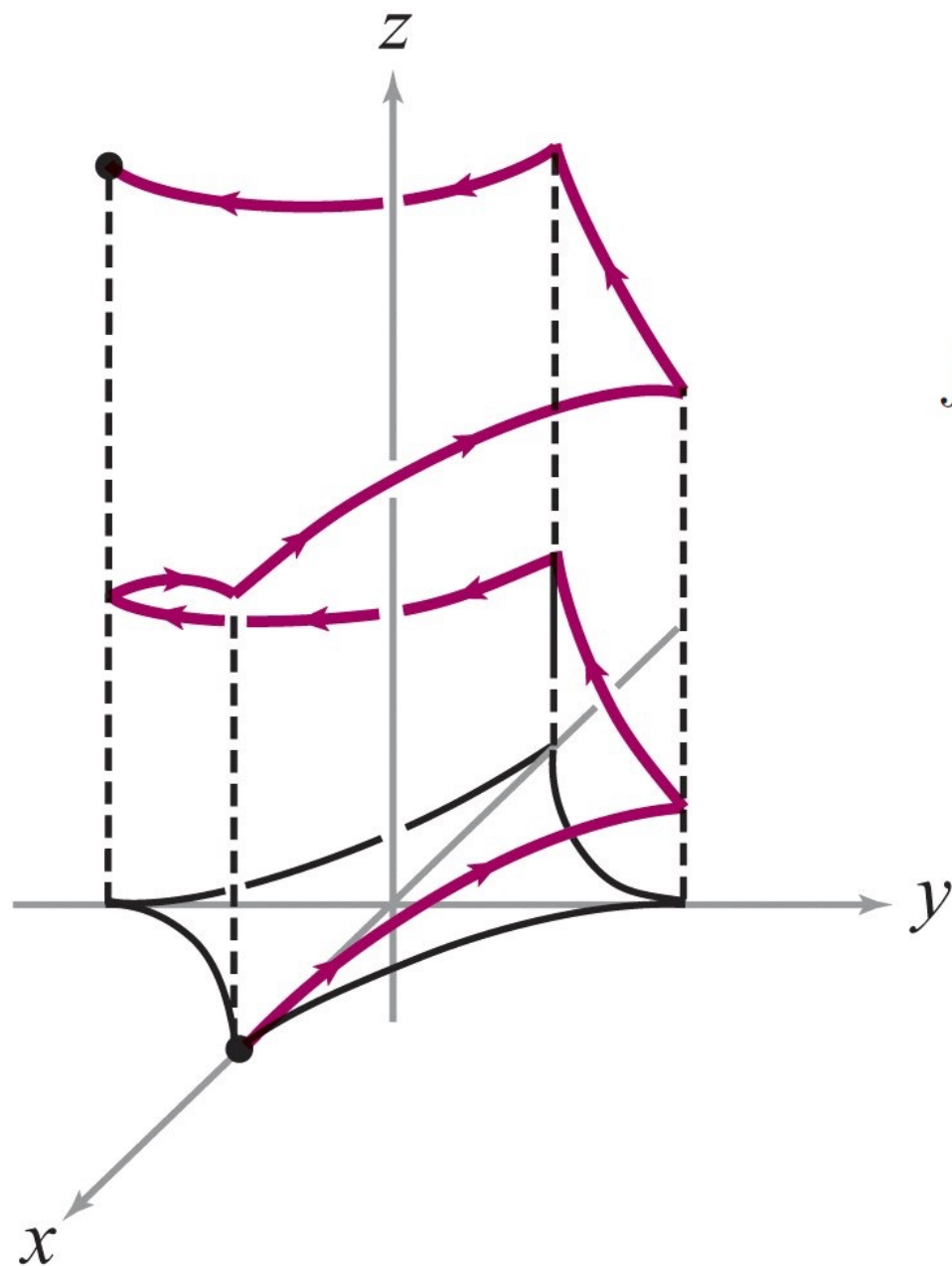
i

$$\mathbf{c}(t) = (x(t), y(t), z(t)), \quad t \in [a, b]$$

aleshores

$$\begin{aligned} \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} &= \int_{\mathbf{c}} F_1 dx + F_2 dy + F_3 dz \\ &= \int_a^b \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt \end{aligned}$$

$$x = \cos^3 \theta, \quad y = \sin^3 \theta, \quad z = \theta, \quad 0 \leq \theta \leq \frac{7\pi}{2}$$



$$\int_c \sin z \, dx + \cos z \, dy - (xy)^{1/3} \, dz$$

$$x = \cos^3 \theta, \quad y = \sin^3 \theta, \quad z = \theta, \quad 0 \leq \theta \leq \frac{7\pi}{2}$$

In this case, we have

$$\frac{dx}{d\theta} = -3 \cos^2 \theta \sin \theta, \quad \frac{dy}{d\theta} = 3 \sin^2 \theta \cos \theta, \quad \frac{dz}{d\theta} = 1,$$

so the integral is

$$\begin{aligned} & \int_{\mathbf{c}} \sin z \, dx + \cos z \, dy - (xy)^{1/3} \, dz \\ &= \int_0^{7\pi/2} (-3 \cos^2 \theta \sin^2 \theta + 3 \sin^2 \theta \cos^2 \theta - \cos \theta \sin \theta) \, d\theta. \end{aligned}$$

The first two terms cancel, and so we get

$$- \int_0^{7\pi/2} \cos \theta \sin \theta \, d\theta = - \left[\frac{1}{2} \sin^2 \theta \right]_0^{7\pi/2} = -\frac{1}{2}.$$

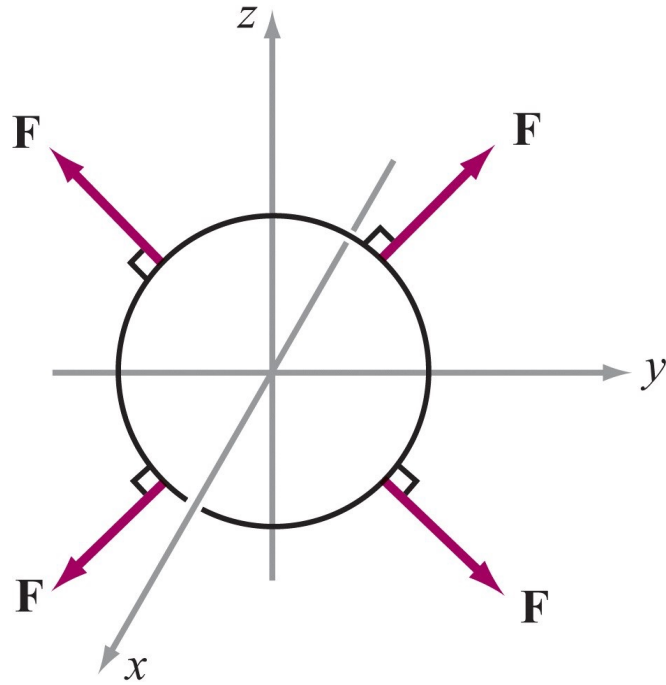
$$\mathbf{F}(x, y, z) = x^3 \mathbf{i} + y \mathbf{j} + z \mathbf{k}$$

$$x = 0, \quad y = a \cos \theta, \quad z = a \sin \theta, \quad 0 \leq \theta \leq 2\pi.$$

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$$

$$\mathbf{F}(x, y, z) = x^3 \mathbf{i} + y \mathbf{j} + z \mathbf{k}$$

$$x = 0, \quad y = a \cos \theta, \quad z = a \sin \theta, \quad 0 \leq \theta \leq 2\pi.$$



$$\mathbf{F}(\mathbf{c}(\theta)) \cdot \mathbf{c}'(\theta) = 0$$

$$\begin{aligned} W &= \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} x^3 dx + y dy + z dz \\ &= \int_0^{2\pi} (0 - a^2 \cos \theta \sin \theta + a^2 \cos \theta \sin \theta) d\theta = 0 \end{aligned}$$

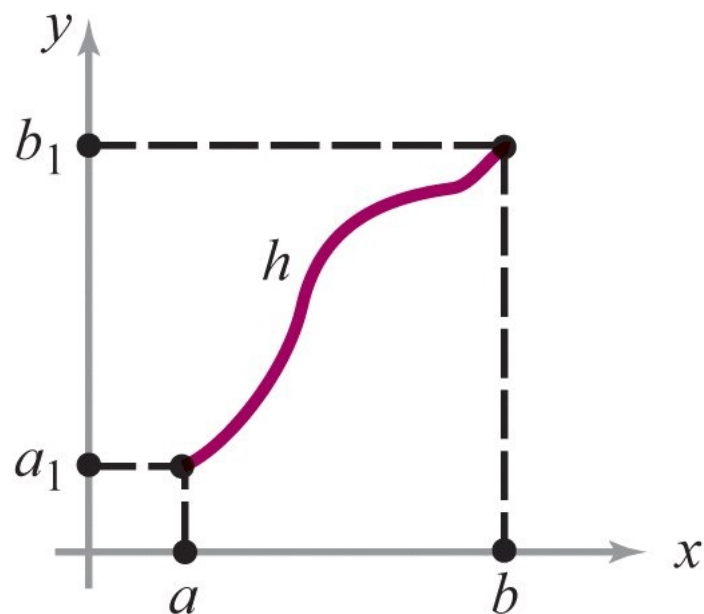
Canvi de variables en integrals de línia

DEFINITION Let $h: I \rightarrow I_1$ be a C^1 real-valued function that is a one-to-one map of an interval $I = [a, b]$ onto another interval $I_1 = [a_1, b_1]$. Let $\mathbf{c}: I_1 \rightarrow \mathbb{R}^3$ be a piecewise C^1 path. Then we call the composition

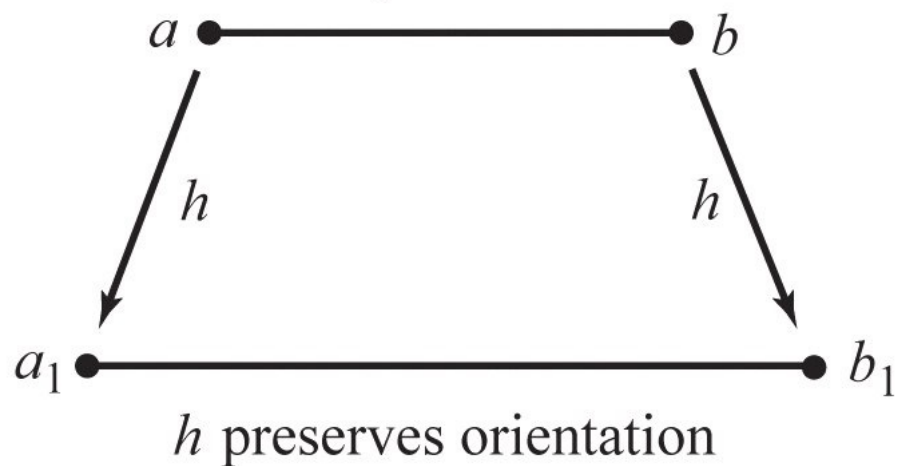
$$\mathbf{p} = \mathbf{c} \circ h: I \rightarrow \mathbb{R}^3$$

a *reparametrization* of $\mathbf{c}$.

Orientation-preserving reparametrization



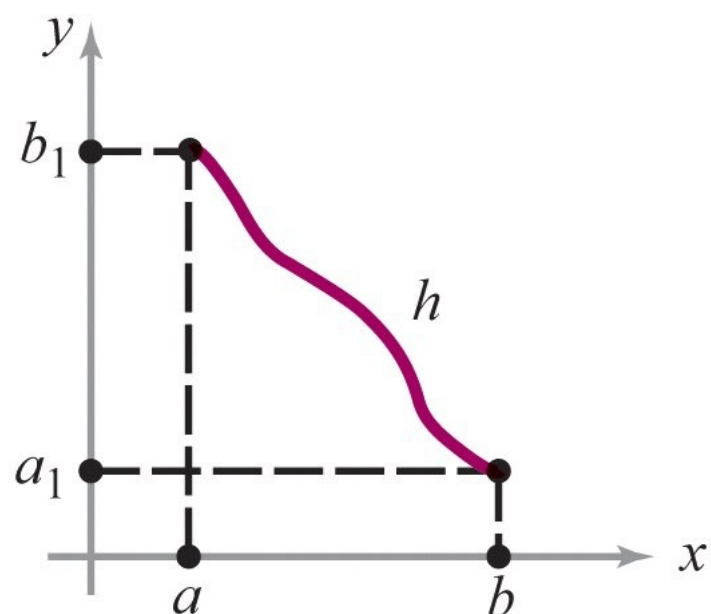
Graph of h



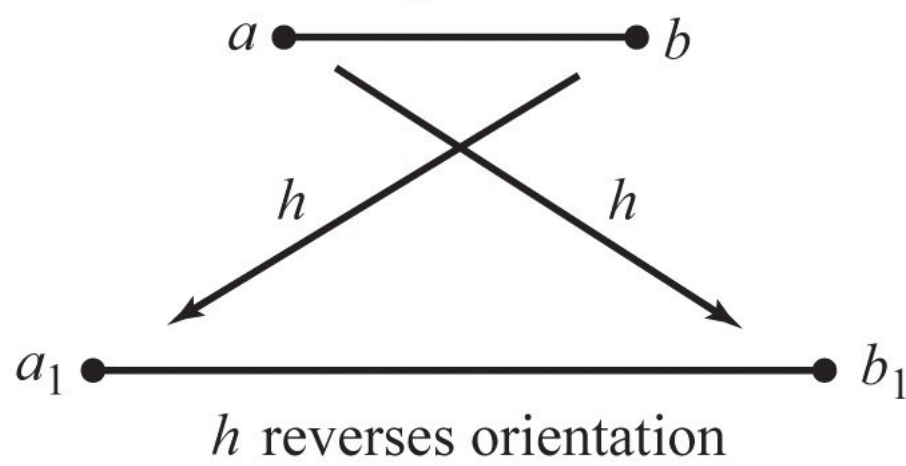
h preserves orientation

(a)

Orientation-reversing reparametrization

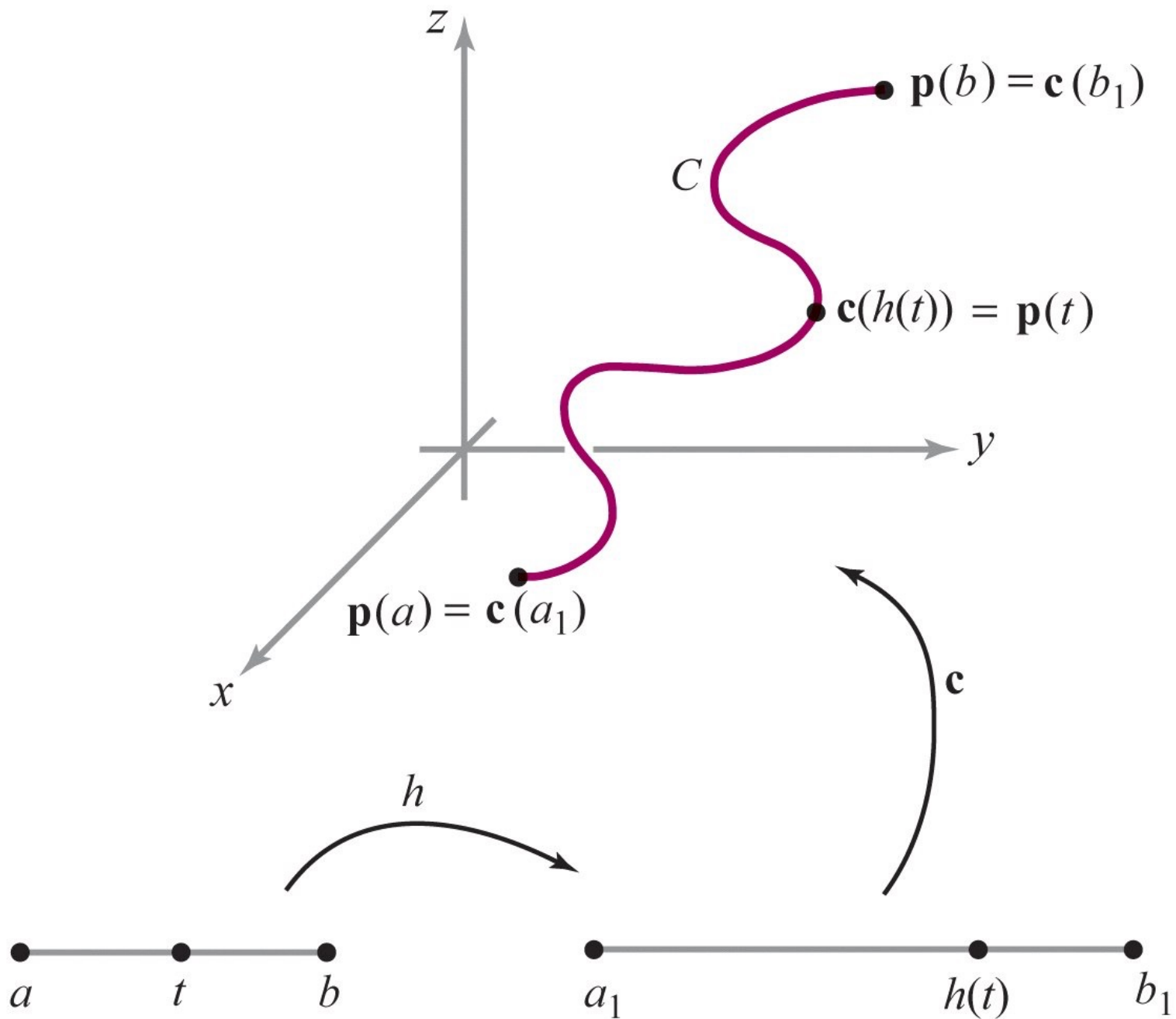


Graph of h



h reverses orientation

(b)



Canvi de variables en integrals de línia de funcions vectorials

THEOREM 1: Change of Parametrization for Line Integrals

Let $\mathbf{F}$ be a vector field continuous on the C^1 path $\mathbf{c}: [a_1, b_1] \rightarrow \mathbb{R}^3$, and let $\mathbf{p}: [a, b] \rightarrow \mathbb{R}^3$ be a reparametrization of $\mathbf{c}$. If $\mathbf{p}$ is orientation-preserving, then

$$\int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s},$$

and if $\mathbf{p}$ is orientation-reversing, then

$$\int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = - \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}.$$

Canvi de variables en integrals de línia de funcions escalars

THEOREM 2: Change of Parametrization for Path Integrals

Let $\mathbf{c}$ be piecewise C^1 , let f be a continuous (real-valued) function on the image of $\mathbf{c}$, and let $\mathbf{p}$ be any reparametrization of $\mathbf{c}$. Then

$$\int_{\mathbf{c}} f(x, y, z) ds = \int_{\mathbf{p}} f(x, y, z) ds. \quad (2)$$

Let $\mathbf{F}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$ and $\mathbf{c}: [-5, 10] \rightarrow \mathbb{R}^3$ be defined by $t \mapsto (t, t^2, t^3)$. Evaluate $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ and $\int_{\mathbf{c}_{\text{op}}} \mathbf{F} \cdot d\mathbf{s}$.

For the path $\mathbf{c}$, we have $dx/dt = 1$, $dy/dt = 2t$, $dz/dt = 3t^2$, and $\mathbf{F}(\mathbf{c}(t)) = t^5\mathbf{i} + t^4\mathbf{j} + t^3\mathbf{k}$. Therefore,

$$\begin{aligned} \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} &= \int_{-5}^{10} \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt \\ &= \int_{-5}^{10} (t^5 + 2t^5 + 3t^5) dt = [t^6]_{-5}^{10} = 984,375. \end{aligned}$$

On the other hand, for $h(t) = 5 - t$, $h(-5) = 10$, $h(10) = -5$

$$\mathbf{c}_{\text{op}}: [-5, 10] \rightarrow \mathbb{R}^3, t \mapsto \mathbf{c}(5 - t) = (5 - t, (5 - t)^2, (5 - t)^3),$$

we have $dx/dt = -1$, $dy/dt = -10 + 2t = -2(5 - t)$, $dz/dt = -75 + 30t - 3t^2 = -3(5 - t)^2$, and $\mathbf{F}(\mathbf{c}_{\text{op}}(t)) = (5 - t)^5\mathbf{i} + (5 - t)^4\mathbf{j} + (5 - t)^3\mathbf{k}$. Therefore,

$$\int_{\mathbf{c}_{\text{op}}} \mathbf{F} \cdot d\mathbf{s} = \int_{-5}^{10} [-(5 - t)^5 - 2(5 - t)^5 - 3(5 - t)^5] dt = [(5 - t)^6]_{-5}^{10} = -984,375. \quad \blacktriangle$$

THEOREM 3: Line Integrals of Gradient Vector Fields Suppose that $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ is of class C^1 and that $\mathbf{c}: [a, b] \rightarrow \mathbb{R}^3$ is a piecewise C^1 path. Then

$$\int_{\mathbf{c}} \nabla f \cdot d\mathbf{s} = f(\mathbf{c}(b)) - f(\mathbf{c}(a)).$$

Això és una generalització del Teorema Fonamental del Càlcul

$$\int_a^b g(x)dx = G(b) - G(a) \quad \text{or} \quad \int_a^b f'(x)dx = f(b) - f(a)$$

proof Apply the chain rule to the composite function

$$F: t \mapsto f(\mathbf{c}(t))$$

to obtain

$$F'(t) = (f \circ \mathbf{c})'(t) = \nabla f(\mathbf{c}(t)) \cdot \mathbf{c}'(t).$$

The function F is a real-valued function of the variable t , and so, by the fundamental theorem of single-variable calculus,

$$\int_a^b F'(t) \, dt = F(b) - F(a) = f(\mathbf{c}(b)) - f(\mathbf{c}(a)).$$

Therefore,

$$\begin{aligned} \int_{\mathbf{c}} \nabla f \cdot d\mathbf{s} &= \int_a^b \nabla f(\mathbf{c}(t)) \cdot \mathbf{c}'(t) \, dt = \int_a^b F'(t) \, dt = F(b) - F(a) \\ &= f(\mathbf{c}(b)) - f(\mathbf{c}(a)). \quad \blacksquare \end{aligned}$$

Exemple

Let $\mathbf{c}$ be the path $\mathbf{c}(t) = (t^4/4, \sin^3(t\pi/2), 0)$, $t \in [0, 1]$. Evaluate

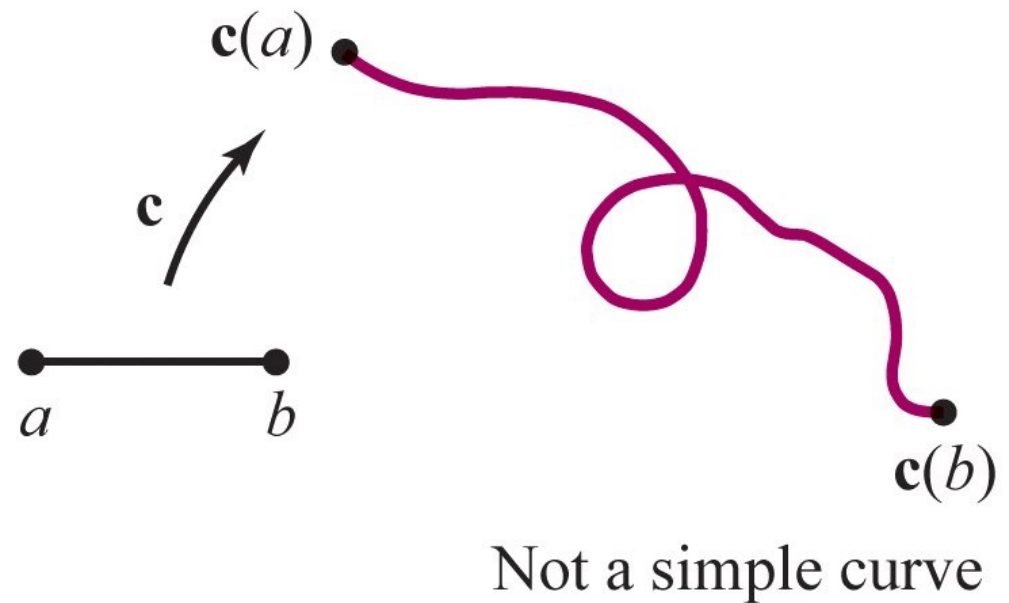
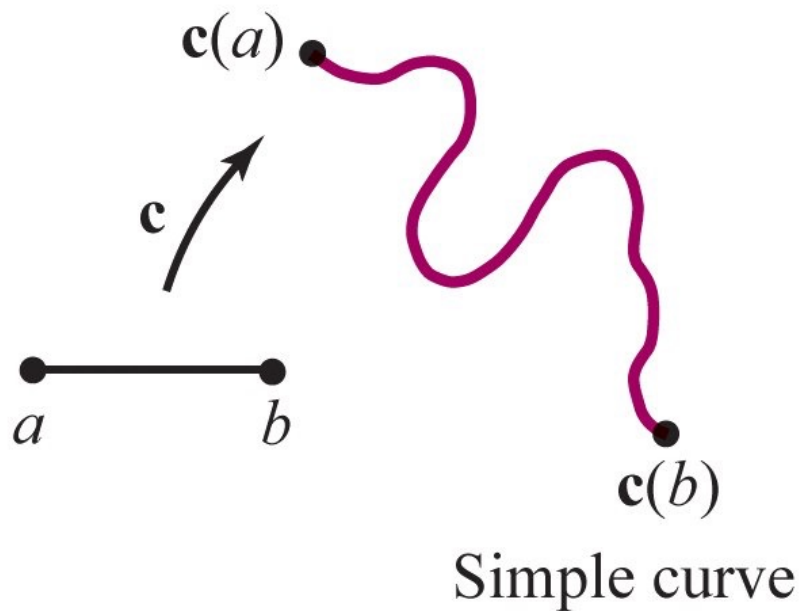
$$\int_{\mathbf{c}} y \, dx + x \, dy$$

(which means $\int_{\mathbf{c}} y \, dx + x \, dy + 0 \, dz$).

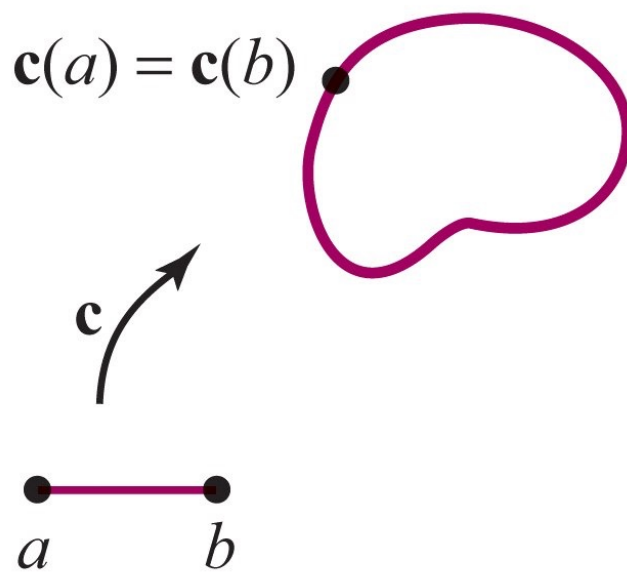
We recognize $y \, dx + x \, dy$, or, equivalently, the vector field $y\mathbf{i} + x\mathbf{j} + 0\mathbf{k}$, as the gradient of the function $f(x, y, z) = xy$. Thus,

$$\int_{\mathbf{c}} y \, dx + x \, dy = f(\mathbf{c}(1)) - f(\mathbf{c}(0)) = \frac{1}{4} \cdot 1 - 0 = \frac{1}{4}. \quad \blacktriangle$$

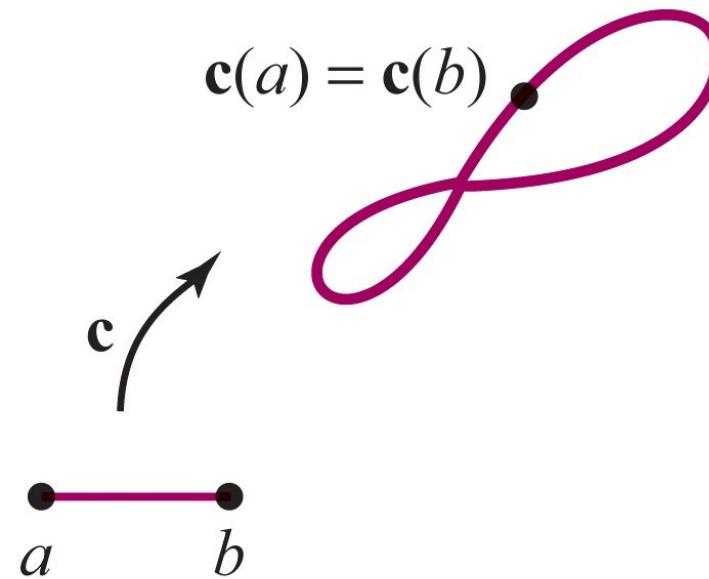
DEFINITION We define a *simple curve* C to be the image of a piecewise C^1 map $\mathbf{c}: I \rightarrow \mathbb{R}^3$ that is one-to-one on an interval I ; $\mathbf{c}$ is called a *parametrization* of C . Thus, a simple curve is one that does not intersect itself (Figure 7.2.6). If $I = [a, b]$, we call $\mathbf{c}(a)$ and $\mathbf{c}(b)$ *endpoints* of the curve.



DEFINITION: Simple Closed Curves By a *simple closed curve* we mean the image of a piecewise C^1 map $\mathbf{c}: [a, b] \rightarrow \mathbb{R}^3$ that is one-to-one on $[a, b)$ and satisfies $\mathbf{c}(a) = \mathbf{c}(b)$ (Figure 7.2.8). If $\mathbf{c}$ satisfies the condition $\mathbf{c}(a) = \mathbf{c}(b)$, but is not necessarily one-to-one on $[a, b)$, we call its image a *closed curve*. Simple closed curves have two orientations, corresponding to the two possible directions of motion along the curve (Figure 7.2.9).

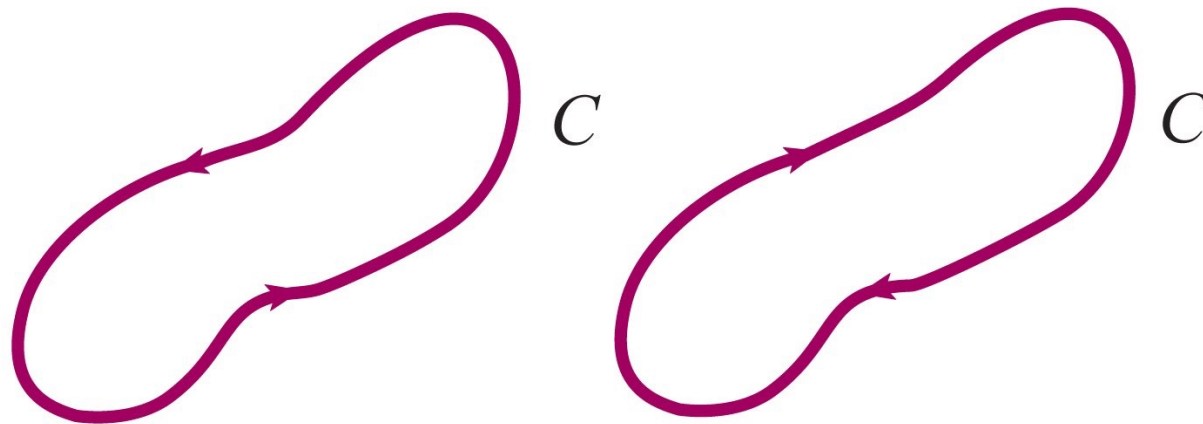


Simple closed curve



Not a simple closed curve

DEFINITION: Simple Closed Curves By a *simple closed curve* we mean the image of a piecewise C^1 map $\mathbf{c}: [a, b] \rightarrow \mathbb{R}^3$ that is one-to-one on $[a, b)$ and satisfies $\mathbf{c}(a) = \mathbf{c}(b)$ (Figure 7.2.8). If $\mathbf{c}$ satisfies the condition $\mathbf{c}(a) = \mathbf{c}(b)$, but is not necessarily one-to-one on $[a, b)$, we call its image a *closed curve*. Simple closed curves have two orientations, corresponding to the two possible directions of motion along the curve (Figure 7.2.9).



The two orientations of a simple closed curve

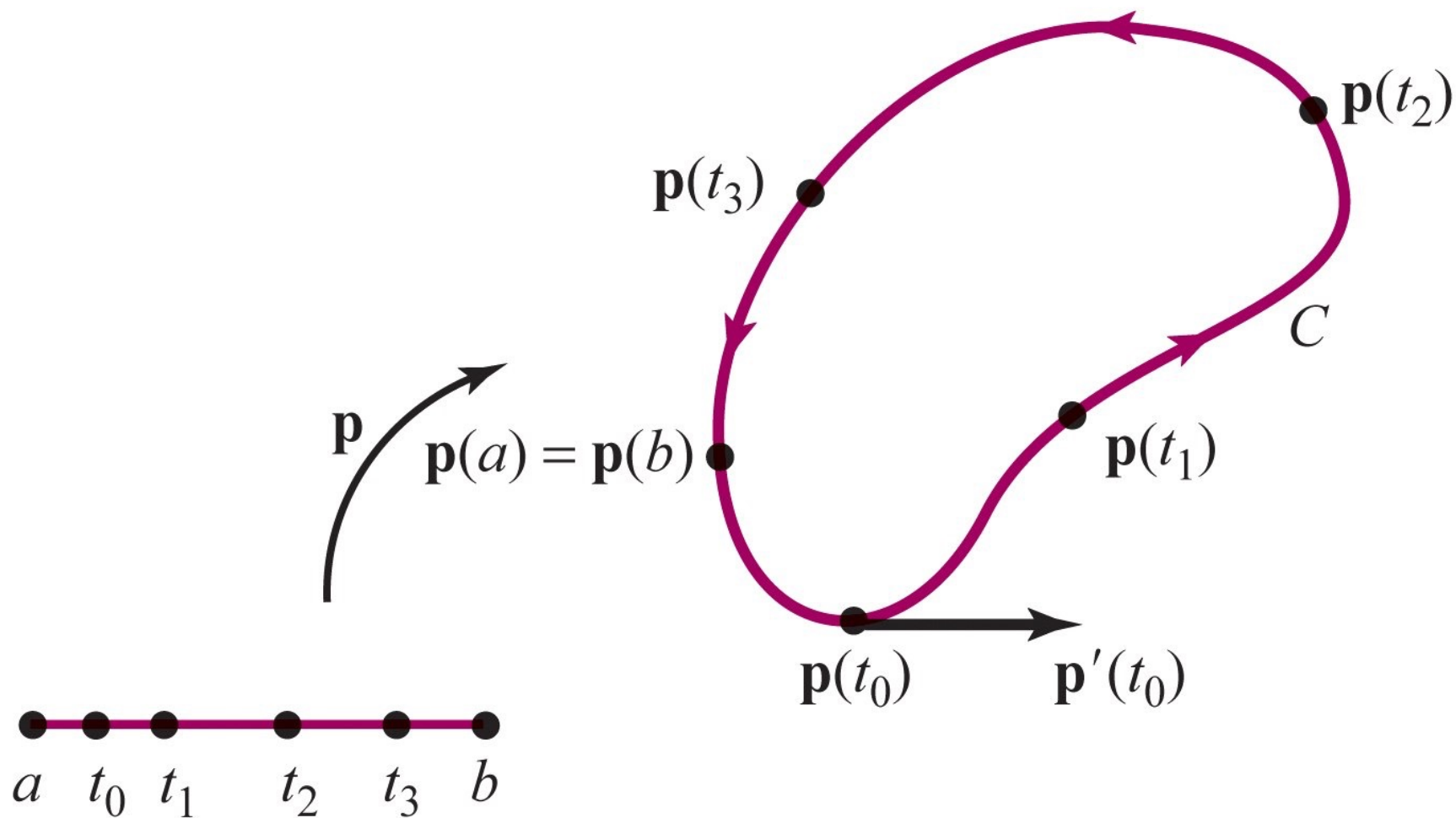
Line Integrals Over Oriented Simple Curves and Simple Closed Curves C :

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} \quad \text{and} \quad \int_C f ds = \int_{\mathbf{c}} f ds, \quad (3)$$

where $\mathbf{c}$ is any *orientation-preserving* parametrization of C .

Això significa que, donada una corba geomètrica (el conjunt de punts), la integral de línia no depèn de la seva parametrització, sempre que preservi l'orientació.

Recordatori: les parametritzacions de corbes simples han de ser bijectives excepte, com a molt, als extrems. Per tant, en aquest teorema s'exclouen les corbes no simples, i les funcions $p: I \rightarrow C$ sobre corbes simples tancades tals que recorren la corba dues o més voltes (p no és parametrització).



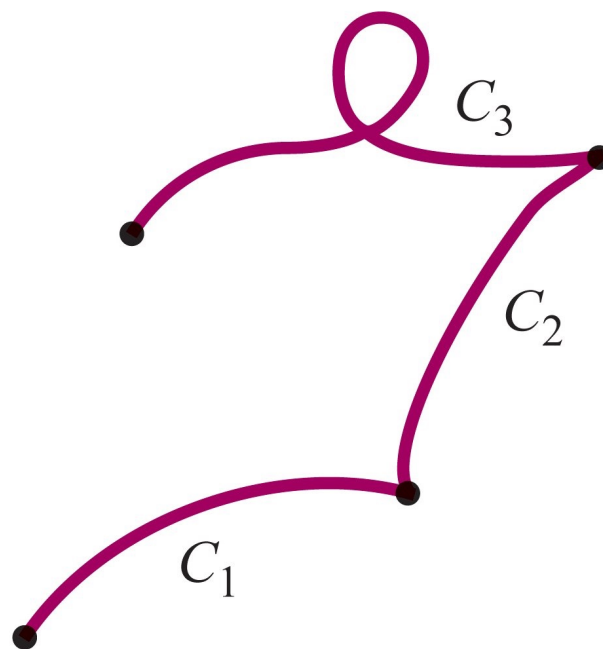
Els vector tangents apunten en la direcció de moviment sobre la corba

Line Integrals Over Curves with Opposite Orientations Let C^- be the same curve as C , but with the opposite orientation. Then

$$\int_C \mathbf{F} \cdot d\mathbf{s} = - \int_{C^-} \mathbf{F} \cdot d\mathbf{s}.$$

Line Integrals Over Curves Consisting of Several Components Let C be an oriented curve that is made up of several oriented component curves C_i , $i = 1, \dots, k$, as in Figure 7.2.11. Then we shall write $C = C_1 + C_2 + \dots + C_k$. Because we can parametrize C by parametrizing the pieces $C_1, \dots, C_k$ separately, one can prove that

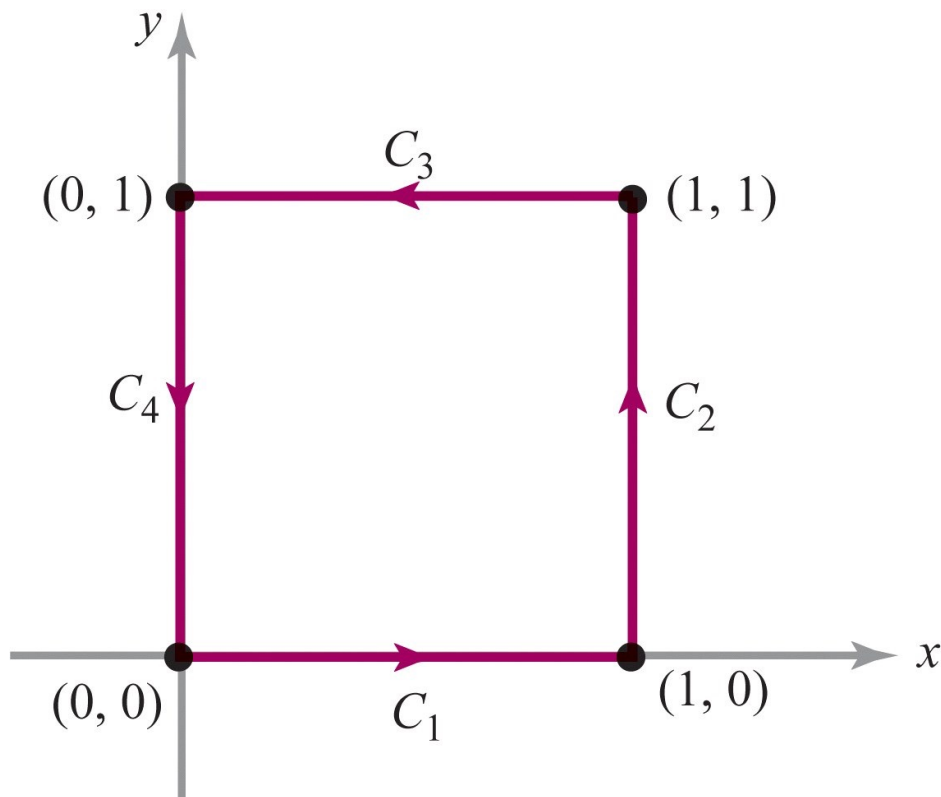
$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{C_1} \mathbf{F} \cdot d\mathbf{s} + \int_{C_2} \mathbf{F} \cdot d\mathbf{s} + \dots + \int_{C_k} \mathbf{F} \cdot d\mathbf{s}. \quad (4)$$



$$\int_C x^2 dx + xy dy = \frac{1}{2}$$

$$\mathbf{c}: [0, 4] \rightarrow \mathbb{R}^2,$$

$$t \mapsto \begin{cases} (t, 0) & 0 \leq t \leq 1 \\ (1, t-1) & 1 \leq t \leq 2 \\ (3-t, 1) & 2 \leq t \leq 3 \\ (0, 4-t) & 3 \leq t \leq 4 \end{cases}$$



$$C = C_1 + C_2 + C_3 + C_4$$

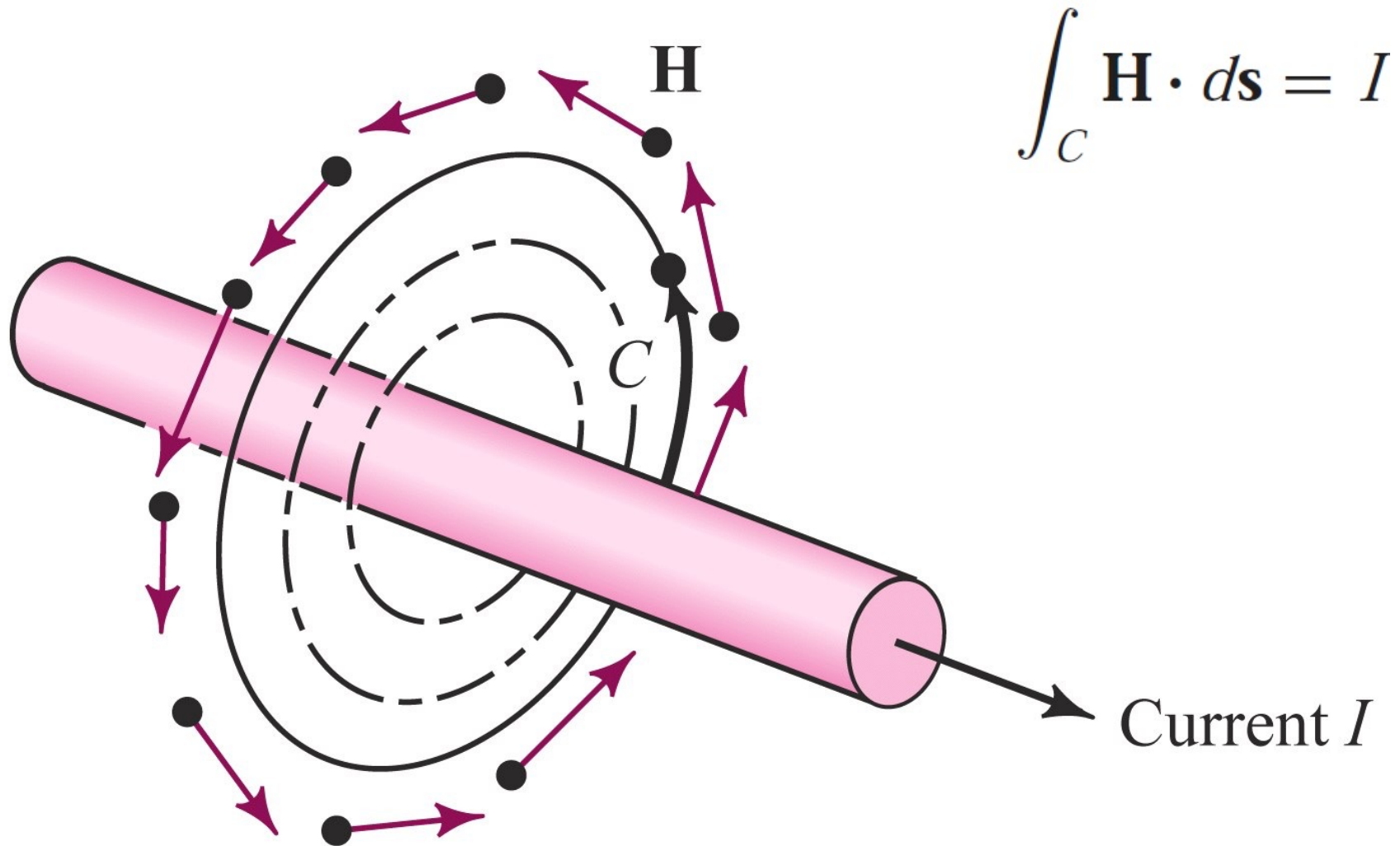
$$C_1: \mathbf{c}_1(t) = (t, 0), 0 \leq t \leq 1$$

$$C_2: \mathbf{c}_2(t) = (1, t), 0 \leq t \leq 1$$

$$C_3: \mathbf{c}_3(t) = (1-t, 1), 0 \leq t \leq 1$$

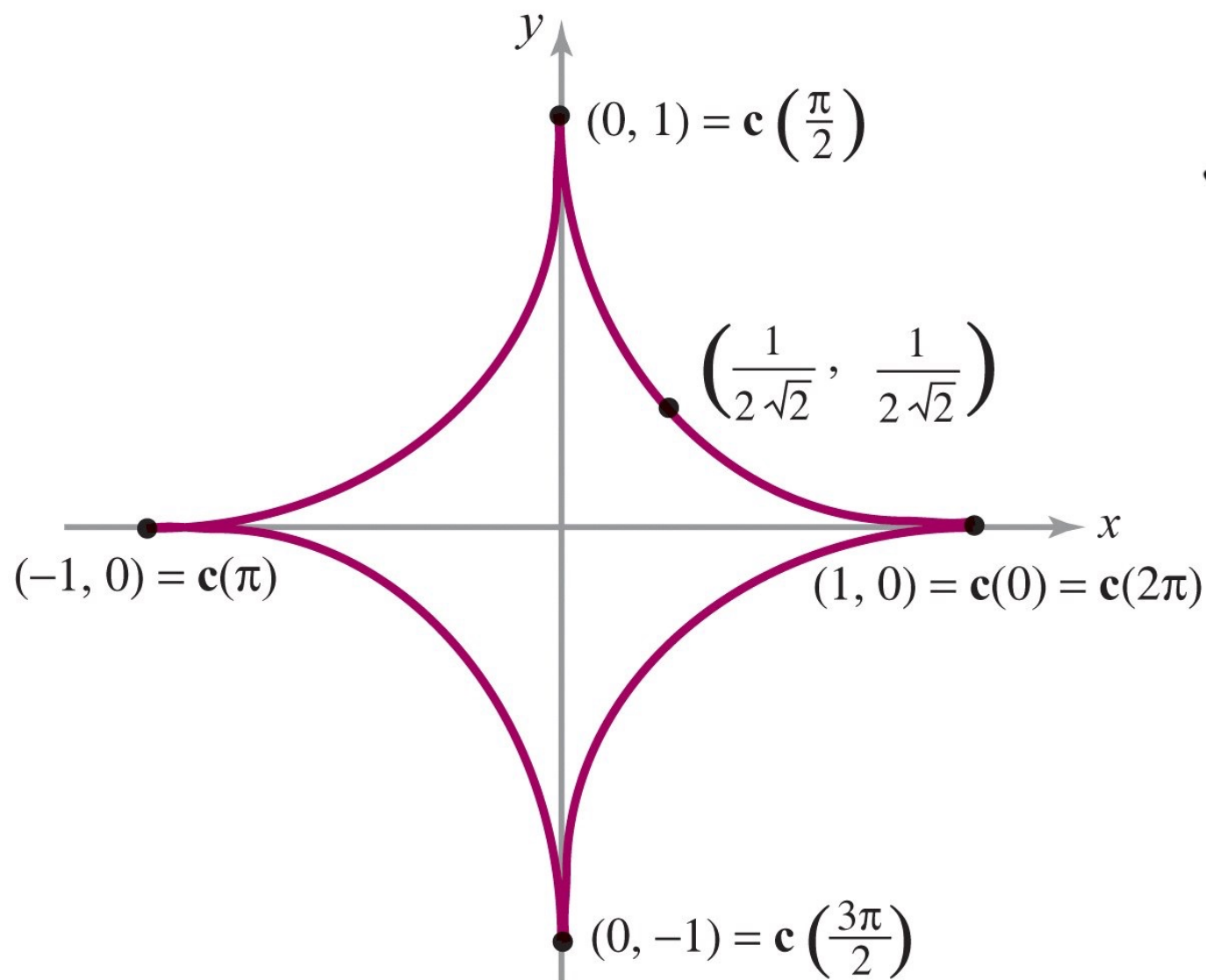
$$C_4: \mathbf{c}_4(t) = (0, 1-t), 0 \leq t \leq 1$$

Llei d'Ampère



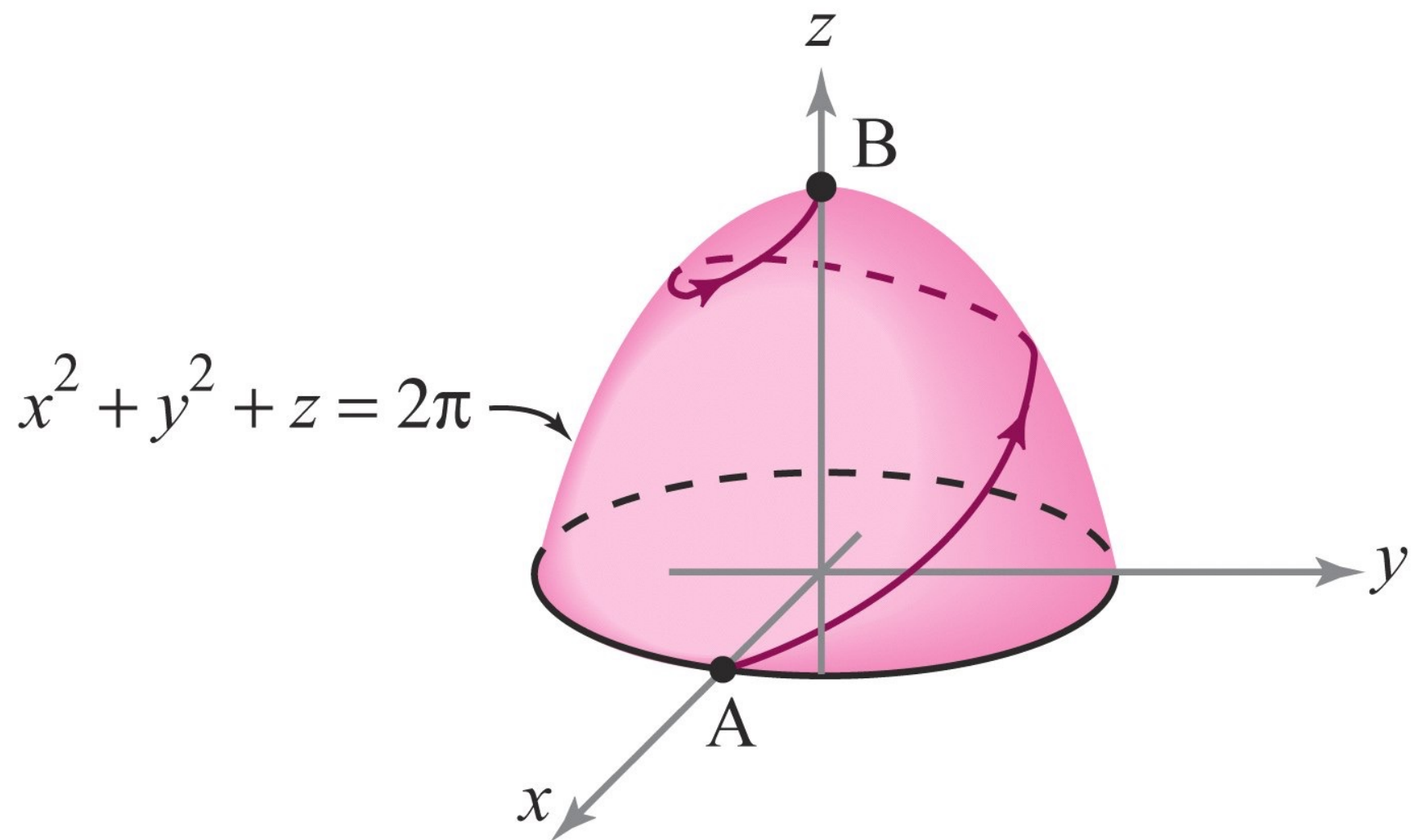
$$\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$$

$$t \mapsto (\cos^3 t, \sin^3 t), 0 \leq t \leq 2\pi$$



$$\int_C \mathbf{F} \cdot d\mathbf{s}$$

Calcula el treball que fa la ciclista per arribar al cim de la muntanya. La velocitat en que incrementa l'alçada és constant.



CONCEPTUAL INSIGHT In physics, two important conservative force fields $\mathbf{F}$ are the gravitational force and electrostatic force. It is customary to change sign and write $\mathbf{F} = -\nabla\varphi$ and to define the **potential energy** (PE) of an object at location (x, y, z) to be $\varphi(x, y, z)$. The work required to move an object from point P to point Q along any path $\mathbf{c}$ is equal to the change in potential energy:

$$\text{Work against } \mathbf{F} = - \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \underbrace{\varphi(Q) - \varphi(P)}_{\text{Change in potential energy}}$$

The name “conservative field” stems from the Law of Conservation of Energy. Suppose that an object subject to a conservative force $\mathbf{F}$ travels along a path $\mathbf{c}(t)$ with velocity $\mathbf{v} = \mathbf{c}'(t)$. By definition, the object’s **kinetic energy** (KE) is $\frac{1}{2}m\mathbf{v} \cdot \mathbf{v}$ and its total energy at time t is the sum

$$E = KE + PE = \frac{1}{2}m\mathbf{v} \cdot \mathbf{v} + \varphi(\mathbf{c}(t))$$

The Law of Conservation of Energy states that E is constant in time. To prove it, we use Newton’s Law $\mathbf{F} = m\mathbf{a}$ (where $\mathbf{a} = \mathbf{v}'(t)$):

$$\begin{aligned} \frac{dE}{dt} &= \frac{d}{dt} \left(\frac{1}{2}m\mathbf{v} \cdot \mathbf{v} + \varphi(\mathbf{c}(t)) \right) \\ &= m\mathbf{v} \cdot \mathbf{a} + \nabla\varphi(\mathbf{c}(t)) \cdot \mathbf{c}'(t) && \text{(by the Chain Rule)} \\ &= m\mathbf{v} \cdot \mathbf{a} + \nabla\varphi(\mathbf{c}(t)) \cdot \mathbf{v} \\ &= m\mathbf{v} \cdot \mathbf{a} - \mathbf{F} \cdot \mathbf{v} && \text{(since } \mathbf{F} = -\nabla\varphi) \\ &= \mathbf{v} \cdot (m\mathbf{a} - \mathbf{F}) = 0 && \text{(since } \mathbf{F} = m\mathbf{a}) \end{aligned}$$